



Literature Review on

LLM DECISION MAKING FOR UAV SIMULATION

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Layout

- **Problem:** Existing UAV navigation methods often separate global planning, local obstacle avoidance, and mission reasoning, making adaptation to dynamic environments difficult and causing unnecessary replanning.
- **Proposed Solution:** An event-triggered hierarchical AI framework using an LLM/VLM for mission-level global planning and a local AI model for real-time trajectory generation and obstacle avoidance.
- **Key Architecture:** The mission planner generates global waypoints; the local navigator follows them while monitoring nearby obstacles and environmental changes.
- **Mechanism:** Continue when locally avoidable; trigger local replanning for temporary obstacles; trigger mission-level replanning when the route is persistently blocked.
- **Application/Goal:** Safe and adaptive autonomous UAV navigation for tasks such as payload delivery, search-and-rescue, and data collection, while reducing unnecessary global replanning.

Problem

- **Difficulty in dynamic environments:** When the environment changes or new obstacles appear, a pre-planned path may no longer be valid, requiring the UAV to adapt during flight.
- **Planning limitation:** Current approaches may trigger unnecessary global replanning even when an obstacle could be handled locally, increasing computation and response time.
- **Reactive limitation:** Local obstacle avoidance can react quickly but may lack high-level mission context and long-horizon reasoning.
- **LLM limitation:** LLMs provide semantic reasoning but need grounding through feasible actions and environmental feedback rather than direct low-level control.

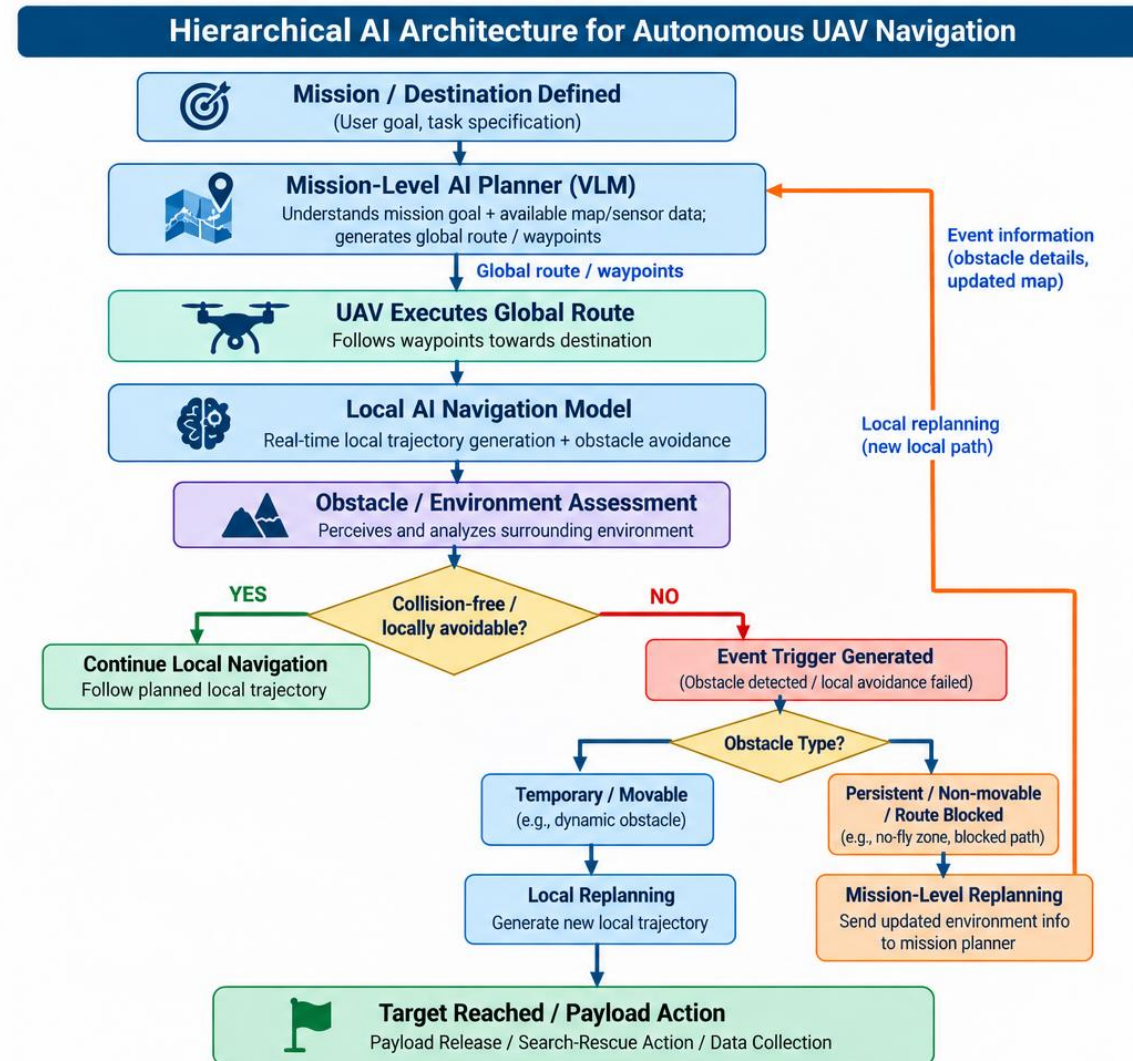
Literature Review

Methodology	Success Rate	Path Length	Collision Rate	Time for Decision-taking	Navigation Error
UAV-VLA (VLM + GPT)	92%	+21.6% vs. human	≈0*	5 min 24 s	34.2 m
VLM-RRT (VLM + RRT)	94%	46 m	6%	89 iterations (RRT: 343)	12 m
FM-Planner (LLM + vision)	100%	14.3 m	≈0%	1 min 20 s	11 m
Standard PPO	~80.6%	Baseline (+27 m)	~19.4%	~181,500 steps	Baseline (>3 m)
SkyAgent (LLM + PPO)*	96.7%	27 m shorter than PPO	83% fewer than PPO	55k steps (3.3× faster than PPO)	3 m
AirVLN-R1 (VLM, fine-tuned)	51.82%	SPL 50.66	~11% missed stops*	0.85 s/step	39.8 m

Proposed Solution

- **Mission-level reasoning:** An VLM interprets the user mission, destination, and available environmental information and produces a global route or waypoint sequence.
- **Local navigation:** A dedicated local AI/navigation module converts the global guidance into safe, real-time trajectories while avoiding nearby obstacles.
- **Event-triggered replanning:** Environmental events determine whether to continue, locally replan, or return to the mission planner for a new global route.
- **Closed-loop feedback:** Local perception and obstacle events update the high-level planner only when the current route becomes unsuitable.
- **Design principle:** Use the LLM/VLM for high-level reasoning, not direct flight control; keep safety-critical local navigation explicit and responsive.

Key Architecture



Research Gap & Proposed Novelty

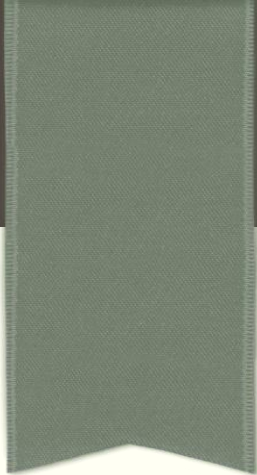
- **Existing work:** LLM/VLM systems demonstrate semantic reasoning and language-guided robotics, while UAV planners address global planning and local obstacle avoidance separately. [1][2][4][5][6]
- **Gap:** These capabilities are not yet sufficiently unified around a UAV-specific decision mechanism that selects the appropriate replanning level after an environmental event.
- **Proposed novelty:** An event-triggered hierarchical architecture that couples mission-level semantic planning with real-time local navigation and selective local/global replanning.
- **Key hypothesis:** Selective replanning can reduce unnecessary global recomputation while maintaining safe navigation and mission completion in dynamic environments.
- **Validation:** Compare against global-only, reactive/local-only, fixed-replanning, and non-LLM hierarchical baselines using success rate, collisions, path length, replanning count, latency, and computation.

References

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THANK YOU